FRI Natural language processing course 2026
Project report

University of Ljubljana
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Natural language processing course project report

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Abstract

We built an AI assistant for the UL FRI student office that answers administrative questions in Slovenian and English. The system is a RAG pipeline that crawls and parses UL/FRI web content, chunks it with a section-aware splitter, embeds the chunks with a multilingual sentence-transformer, and stores them in FAISS. At query time we compare three retrieval modes – dense, BM25+dense hybrid, and hybrid with a cross-encoder reranker – feeding an instruction-tuned generator. We evaluate the pipeline with both classical retrieval metrics (Recall@k, MRR@k, nDCG@k) and a RAGAS-style LLM-as-judge protocol scoring faithfulness, hallucination, refusal handling, and overall quality. Early on we noticed that many of our reference answers were not actually grounded in the corpus, so we first ran a Sonnet-graded audit and used the cleaned set for every comparison. An embedder $\times$ retrieval-mode sweep followed by a chunk-size sweep picks `multilingual-e5-base` with hybrid retrieval, a cross-encoder reranker, top-k=4, and 400-token chunks as the best configuration, with a Sonnet-judged reference-grounding ceiling of 0.748 in-scope faithfulness on the cleaned set. A three-way cluster comparison of the actual generators on this fixed retrieval setup (`GaMS3-12B`, `Llama-3.1-8B`, `Mistral-7B`) makes `Llama-3.1-8B-Instruct` the best generator at 0.665 faithfulness with 7/10 correct refusals on out-of-corpus probes.

Keywords

Retrieval augmented generation, dense retrieval, hybrid retrieval, multilingual embeddings, student affairs chatbot

Introduction

We are building an AI assistant for the student office of the Faculty of Computer and Information Science (UL FRI). The goal is a chatbot that answers everyday administrative questions about study programmes, regulations, and the office's work in Slovenian and English. Plain LLMs are a poor fit here: the answers depend on small details (deadlines, paragraph numbers, application forms) that need to come from the official documents, not from the model's training data. We therefore build the assistant on Retrieval Augmented Generation [1], which has been a good fit for university-style FAQ chatbots in prior work [2, 3] and which a recent survey shows helps with hallucination and out-of-date knowledge in educational deployments [4]. Our pipeline borrows the comparative-retrieval setup (dense vs. hybrid vs. reranked, with a chunk-size sweep) from RAG-for-admissions and university-resources studies [5, 6], and the choice to evaluate locally hosted instruction-tuned LLMs from work on open-day chatbots [7] and on RAG vs. fine-tuning for knowledge-intensive tasks [8].

Methods

Data collection and parsing

The pipeline is organised as *runs*, each holding raw downloads, parsed JSONL, chunks, and a FAISS index in one directory. A lightweight crawler walks breadth-first from a curated seed list with separate depth limits for `fri.uni-lj.si`, other `*.uni-lj.si` domains, and external domains; it downloads HTML and linked PDF/DOCX, augments with sitemaps and RSS/Atom feeds, respects `robots.txt`, deduplicates by SHA-256, and reuses `ETag`/`If-Modified-Since` on updates. Each document becomes a common JSON record (URL-derived `doc_id`, title, language, ordered sections): HTML via BeautifulSoup + trafilatura, PDF via Docling+Tesseract or PyMuPDF, DOCX via `python-docx`.

Chunking, embeddings and retrieval

We split documents with a section-aware sliding window measured in whitespace tokens (default 400, overlap 80). Every chunk is prefixed with the document title and section name, gets a stable `chunk_id`, and is then deduplicated with a SimHash filter. The chunks are embedded with `intfloat/multilingual-e5-base` (768-dim, E5-style `passage:`/`query:` prefixes, L2 normalisation) and indexed in FAISS as an `IndexFlatIP` (cosine under normalisation). At query time we support three retrieval modes: dense (top-k from FAISS), hybrid (BM25 from `rank_bm25` with default k_1=1.5, b=0.75, fused with dense scores after min–max normalisation on the union of both top lists), and hybrid plus cross-encoder rerank using `mmarco-mMiniLMv2-L12-H384-v1` on a candidate pool of size 20. Retrieved chunks are then formatted into a numbered context block and a language-specific system prompt tells the generator to answer *only* from that block and in the detected language.

Evaluation set and reference audit

Our evaluation set is a hand-written JSONL file of 61 questions (51 *in-scope* – the corpus contains a factual answer – and 10 *negatives*, deliberate out-of-corpus probes used to test refusal), shipped with the repository. Each entry carries the expected language, a list of expected keywords, and a free-form `reference_answer` that we use as the human-written gold for the LLM-as-judge protocol below. A smaller development corpus of 25 questions also carries `relevant_doc_ids` (manually annotated) which we use for the document-level retrieval metrics.

The first time we ran the judge, in-scope faithfulness was very low. Inspection showed many of our reference answers contained details the corpus actually does not support, so the judge was, correctly, marking them down. To stop this reference drift from contaminating later comparisons we run a Sonnet-graded fact-check before the cross-configuration sweeps: for each positive question we retrieve a top-30 BM25 $\cup$ dense pool from the current index and ask Claude Sonnet (`claude-sonnet-4-6`, T=0) to label every claim in the reference as `supported`, `partial`, or `unsupported`, and to optionally produce a grounded rewrite. The application policy is fixed: `supported` keeps the original; `partial` or `unsupported` with a rewrite uses the rewrite (flagged `auto_rewritten`); `unsupported` without a rewrite is dropped; negatives are never modified. The cleaned set is written as a versioned artefact `questions_full_v2.json`.

LLM-as-judge protocol

We use a RAGAS-style judge: for each row we send the question, the retrieved top-k chunks, the model answer, and the reference answer to Claude with a strict-JSON rubric covering *faithfulness*, *answer relevance*, and *context relevance* (each in $[0,1]$), binary *hallucination* (per-answer flag, set when at least one claim is unsupported by the retrieved context) and *refusal* flags, and an integer *overall* score in $[1,5]$. Negatives are scored under the same rubric, with a correct refusal rewarded with high faithfulness. All judge calls run at T=0 for reproducibility. To keep cost down without sacrificing the bottleneck quality we use Sonnet (`claude-sonnet-4-6`) for the reference audit and the final re-judgment, and Haiku (`claude-haiku-4-5`) for the comparison sweeps; the two agreed to within 0.01 in in-scope faithfulness on the winning configuration (0.739 vs. 0.748), so we trust Haiku for ranking.

Per-configuration indices are built by driver scripts that pass `EMBEDDING_MODEL`/`CHUNK_SIZE`/`CHUNK_OVERLAP` as environment overrides and persist them next to the index for safety on load.

Results

Corpus, evaluation set, and reference audit

At the default 400/80 chunk policy the full crawl yields 2,742 parsed documents and 29,296 chunks, up from the 176 documents in the initial *Test* corpus we used during development. Building the index for the full crawl took roughly 4 hours when OCR was on; disabling OCR (PyMuPDF text extraction only) brought that down to $\sim$ 14 minutes for the same $\sim$ 500 PDFs and we did not see any measurable retrieval-quality drop afterwards, so we kept it off for all later experiments. Running the Sonnet audit on the original references against this corpus was a wake-up call: only 9/47 ($\sim$ 19%) of the in-scope references were fully supported as written. Of the rest, 31 were classified `partial` and rewritten using only chunk-attested facts, 7 were `unsupported` (3 dropped, 4 rewritten where Sonnet found alternative chunks), and 14 were skipped (no factual reference, or negative). The cleaned `questions_full_v2.json` contains 51 in-scope (of which 44 have a written reference answer) and 10 negatives, and is the basis for everything that follows.

Document-level retrieval (development corpus)

On the smaller 25-question development corpus, where we have document-level annotations, dense retrieval at k=4 already reaches 88.0% Recall@4 and 66.3% nDCG@4. Adding the cross-encoder reranker lifts these to 100.0% and 93.4%, while hybrid + rerank is essentially tied (100.0% / 93.9%). The corresponding chunk-size sweep on the same corpus showed sizes 200–500 all reaching $\sim$ 97.9% Recall, with 300 giving the highest precision; the LLM-judged sweep on the full crawl, reported below, revisits this choice on a stricter metric and ends up picking 400.

Embedder $\times$ retrieval-mode sweep (LLM-as-judge)

Table 1 shows the embedder $\times$ retrieval-mode sweep at top-k=4 on the cleaned set, judged by Claude Haiku at T=0; refusal-side faithfulness is ≥ 0.90 across all six configurations. Two things stood out for us. First, the cross-encoder reranker is by far the biggest lever: on `e5-base` it lifts faithfulness from 0.575 (dense) and 0.565 (hybrid) to 0.739, and context relevance from about 0.72 to 0.857. Second, the result we did not expect – `e5-base + rerank` beats `e5-large + rerank` on every in-scope metric while embedding about 2.5$\times$ faster. Once the cross-encoder does the final ranking, the smaller first-stage embedder is enough.

Chunk size on the winning configuration

Holding the winner fixed (`e5-base`, hybrid + rerank, k=4) and varying the chunk size with 20% overlap, the Haiku-judged in-scope faithfulness is 0.691 at 200/40 (50,546

Table 1. Embedder × retrieval-mode sweep on `questions_full_v2.json` (Haiku judge, top-k=4, T=0). **Faithfulness**: fraction of answer claims supported by the retrieved chunks ($[0,1]$, higher is better). **Hallucination**: per-answer rate of at least one unsupported claim (lower is better). **Ctx rel.**: judge-rated relevance of retrieved chunks to the question ($[0,1]$). **Overall**: integer rubric score in $[1,5]$.

Embedder	Mode	Faithfulness	Hallucination	Ctx rel.	Overall (1–5)
`multilingual-e5-base`	dense	0.575	59.1%	0.725	3.000
`multilingual-e5-base`	hybrid	0.565	63.6%	0.715	2.841
`multilingual-e5-base`	**hybrid + rerank**	**0.739**	**54.5%**	**0.857**	**3.432**
`multilingual-e5-large`	dense	0.680	54.5%	0.812	3.250
`multilingual-e5-large`	hybrid	0.650	61.4%	0.787	3.205
`multilingual-e5-large`	hybrid + rerank	0.708	59.1%	0.866	3.364

chunks), 0.698 at 300/60 (36,331), and 0.739 at 400/80 (29,296). Larger chunks help here because the cross-encoder benefits from a bit more local context per chunk, and 400-token chunks also produce the smallest, fastest index of the three – so we keep them as the default.

Reference-grounding ceiling and generator comparison

We re-judged the winning retrieval configuration with Claude Sonnet at T=0 in two ways. First, as a *reference-grounding ceiling*: given the retrieved top-4 chunks, how well-supported are the cleaned reference answers themselves? This is an upper bound on what any extractive generator could achieve under our retrieval. The reference set scores 0.748 in-scope faithfulness, 0.826 answer relevance, 0.810 context relevance, 44.2% hallucination, and 10/10 correctly refused negatives – consistent with the Haiku-judged 0.739, so the two judges agree to within 0.01. Walking the gain through the retrieval stages: the original baseline (small corpus, noisy refs, dense k=8) sat at Haiku-judged faithfulness of 0.34; the full crawl plus reference audit at dense k=4 already moves it to 0.575; hybrid alone does not help (0.565); adding the cross-encoder reranker takes it to 0.739 Haiku / 0.748 Sonnet, with per-answer hallucination dropping from 80.9% to 44.2%.

Second, we ran an actual generator comparison on the cluster (`onj_fri` partition) with the same retrieval, judge, and cleaned set (Table 2). `Llama-3.1-8B-Instruct` comes out on top at 0.665 in-scope faithfulness, 50.0% hallucination, 7/10 correctly refused negatives. `Mistral-7B-Instruct-v0.3` is essentially tied on faithfulness (0.653) and slightly better on context relevance but slower. `GaMS3-12B-Instruct`, despite being trained for Slovenian, came out lowest in faithfulness (0.592) with the highest hallucination rate (66.7%) and only 6/10 correct refusals. The gap between the best generator (0.665) and the reference-grounding ceiling (0.748) is small: under our retrieval and judge, most of the remaining headroom is on the retrieval side rather than the generator side.

Discussion

Both the retrieval-only and the LLM-as-judge results point at the same story. The biggest improvements split roughly evenly between two stages: corpus expansion and the Sonnet audit together move dense-k=4 faithfulness from 0.34 to 0.575 (about 58% of the total gain), and the cross-encoder reranker takes it from there to 0.748 (the remaining 42%). Hybrid retrieval on its own does almost nothing, which surprised us; it only starts to pay off once the reranker is on top of it, presumably because the union of BM25 and dense candidates gives the cross-encoder a more diverse pool to choose from. Another finding we did not expect was that `multilingual-e5-base` beats `multilingual-e5-large` once reranking is in play while embedding $\sim 2.5\times$ faster: the cross-encoder absorbs the work that would otherwise depend on a stronger first-stage embedder.

The generator comparison sharpens the picture. The actual best generator (`Llama-3.1-8B-Instruct`, 0.665 faithfulness) leaves only a 0.083 gap to the reference-grounding ceiling (0.748): under our retrieval and judge, more than half of the system's headroom is on the retrieval side, not the generator side. `GaMS3-12B-Instruct` being the weakest of the four also surprised us; it is the largest model and the only one trained specifically for Slovenian, but it hallucinated more often (66.7% per-answer rate) and refused fewer negatives (6/10) than the smaller multilingual baselines. Per-question inspection shows the failure mode: on the negative *"Recept za potico z orehi"*, GaMS produces a recipe rather than refusing, while Llama and Mistral correctly say the corpus has no such information; on the in-scope *"Kdaj poteka vpis v višji letnik"*, Llama answers verbatim from a single chunk (faithfulness 1.00) while GaMS adds details not present in any retrieved chunk (faithfulness 0.55, hallucination flagged). The pattern is consistent: GaMS prefers fluent Slovenian narration over strict context fidelity, which the judge penalises. We did not investigate whether this is prompt-format sensitivity, instruction-tuning data, or something else.

Direct comparison against prior university-RAG work [5, 6, 7] is difficult because their corpora and metrics differ (English admissions FAQs, accuracy on closed-set MCQs, etc.), so absolute numbers are not aligned – this is why we

Table 2. Generator comparison on the cluster (Sonnet judge, $T=0$, same retrieval, top-$k=4$). Same metric definitions as Table 1. n_{is} is the in-scope sample after the judge separates refusals (lower means the generator refused more often). **Refused / 10** counts correctly refused out-of-corpus negatives. The bottom row is the reference-grounding ceiling: Sonnet judging the cleaned references themselves against retrieved chunks; $n=54$ here because 7 in-scope v2 questions added after the audit have no written reference answer.

Generator	n_{is}	Faithfulness	Hallucination	Ctx rel.	Overall	Refused / 10
`GaMS3-12B-Instruct`	54	0.592	66.7%	0.688	2.907	6/10
`Llama-3.1-8B-Instruct`	52	**0.665**	**50.0%**	0.707	**3.000**	**7/10**
`Mistral-7B-Instruct-v0.3`	51	0.653	58.8%	**0.729**	2.961	7/10
Reference-grounding ceiling	43	0.748	44.2%	0.810	3.605	10/10

limit ourselves to within-system comparisons (retrieval ablation, embedder sweep, generator sweep) and report all four under one consistent judge. The biggest caveats are around the evaluation set: 51 in-scope and 10 negative questions are not many, the language is mostly Slovenian, and even with $T=0$ a single LLM judge has some per-question variability – a stratified manual review with Cohen's κ against the Sonnet judge is the next thing on our list. We also disabled OCR to keep full-crawl build time tractable: for scanned PDFs this would silently drop content, so a future iteration should re-run Docling/Tesseract just on the subset where PyMuPDF returns near-empty text.

Code and data availability

The complete pipeline, cleaned evaluation set, FAISS index, all judge/driver scripts, and SLURM launchers are released at the project repository (branch `eval-pipeline`). Seed URLs are in `raw_dataset/data_links.txt`; `raw/` and `parsed.jsonl` are regenerable. We train no model – all embedders, reranker, and generators are open-weight on Hugging Face Hub and the judge is Anthropic's API. Only `ANTHROPIC_API_KEY` and `HF_TOKEN` are needed; `code/QUICKSTART.md` is the reproduction recipe.

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